

46770 Integrated Energy Grids

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Lecture 3 – Roleplay electricity market

Information

System overview

- Technologies
 - Coal
 - Gas
 - Hydropower
 - Onshore wind
 - Offshore wind
 - Solar PV
- Five time periods during which participants can bid
- Every round, you get a weather / status forecast
 - Demand forecast will be announced
 - Availability of technologies will be announced
- Technologies have different marginal prices (e.g. fuel cost)

Participants

- For each technology, we have two participants with 50% of the capacity
 - Coal
 - Gas
 - Hydropower
 - Onshore wind
 - Offshore wind
 - Solar PV
- Hydropower is a special case: it consists of a reservoir (limited energy storage) and pumped hydro storage (which it can decide to charge)
 - Bidding as generator **and** consumer (possibly increasing demand)

Rules

- Every generator submits a bid with price and quantity (cannot bid more than available technology)
- Compound bids are possible: e.g. 100 MW at 30 EUR/MWh, 200 MW at 40 EUR/MWh, 50 MW at 50 EUR/MWh etc.
- The bids are secret and all happen at the same time
- Bids will not be disclosed!

Period 1

Period 1 - 06:00

- Cold weather
 - Fairly low winds (20% capacity factor)
 - Early morning, irradiance at 10%
-
- Demand forecast: 3000 MWh

Submit your bids!

Results

Clearing price: EUR/MWh

Selected generators:

Total demand:

Period 2

Period 2 - 11:00

- Windy (50% capacity factor)
 - Sun is shining (90% capacity factor)
-
- Demand forecast: 4000 MWh

Submit your bids!

Period 3

Period 3 - 17:00

- Solar resource at 20%
- Still conditions (10% wind resources)
- Unexpected shutdown of coal power plants (-500 MW capacity)
- Demand forecast: 4500 MWh

Submit your bids!

Period 4

Period 4 - 20:00

- Evening sun (20%)
 - Windy (80%)
-
- Demand forecast: 5500 MWh

Submit your bids!

Period 5

Period 5 - 04:00

- Night time (no solar)
 - Windy!!!! (100% availability)
-
- Demand forecast: 2000 MWh

Submit your bids!

Settling up

Total score

Generator	Profit	Generator	Profit
Coal 1		Coal 2	
Gas 1		Gas 2	
Hydro 1		Hydro 1	
Onshore wind 1		Onshore wind 2	
Offshore wind 1		Offshore wind 2	
Solar PV 1		Solar PV 2	

Assumptions

Market design / Bidding

Market power / competition

Opportunity cost

Information level: demand, weather, outage, who knows about what?

Compound bidding

Forecast?

How did you bid? Marginal price = bid?

Realism of assumptions?

- Perfect information: capacities, marginal cost, storage level
- Perfect foresight
- Perfect competition: price taking, no market power
- One bid vs compound bidding (open vs secret bids)
- Clearing price = marginal cost of last generator in merit order

DTU



Round 2

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Participants

- For each technology, we have two participants with 50% of the capacity
 - Coal: total capacity 2.4 GW
 - Gas: total capacity 1.1 GW
 - Hydropower: discharge capacity 1 GW, pumping cap. 300 MW, energy cap. 3 GWh
 - Onshore wind: total capacity 3.8 GW
 - Offshore wind: total capacity 1.2 GW
 - Solar PV: total capacity 400 MW
- Hydropower is a special case: it consists of a reservoir (limited energy storage) and pumped hydro storage (which it can decide to charge)
 - Bidding as generator **and** consumer (possibly increasing demand)

Rules

- Every generator submits a bid with price and quantity (cannot bid more than available technology)
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Average bid from before

- Coal 1:
- Coal 2:
- Gas 1:
- Gas 2:
- Hydro 1:
- Hydro 2:
- Onshore wind 1:
- Onshore wind 2:
- Offshore wind 1:
- Offshore wind 2:
- Solar PV 1:
- Solar PV 2:

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Settling up

Total score

Generator	Profit round 1	Profit	Generator	Profit round 1	Profit
Coal 1			Coal 2		
Gas 1			Gas 2		
Hydro 1			Hydro 1		
Onshore wind 1			Onshore wind 2		
Offshore wind 1			Offshore wind 2		
Solar PV 1			Solar PV 2		

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